Training through JPEG without pretending JPEG is differentiable

The forward pass performs exact rounding; the backward pass uses a straight-through gradient. Final claims are re-measured with real files.



training image x

COCO crop

table predictor fθ



thumbnail + DCT stats

predicted tables T

Y Cb Cr

uniform noise ±0.5 for
soft integer-table training
hard rounding at inference

differentiable JPEG

YCbCr · 4:2:0 · 8×8 DCT

forward: round(DCT ÷ T)
backward: identity gradient

IDCT · upsample · RGB

reconstruction x̂JPEG



x̂JPEG

distorted image

mirror autoencoder

T → transposed Conv + IGDN → x̂AE

LAE = MSE(x, x̂AE)

historical training tool; λAE = 0 in finalist

soft table entropy

Rtable = $-\Sigma \log_2 p\psi(T)$

factorized Ballé-style density

optional regularizer; λtable = 0 in finalist

DCT symbol-rate proxy

RDCT = $-\Sigma \log_2 p\text{Laplace}(k)$

separate component × frequency models

training gradient—not reported file size

perceptual distortion

1 – MS-SSIMRGB

+ 0.25 (1 – MS-SSIMY)

optional JPEG MSE guard

L = λd D + 0.05 RDCT + λtable Rtable + λAE LAE + λMSE LMSE

the released continuation uses λd = 1, luma guard = 0.25, λtable = 0, λAE = 0, λMSE = 0

release gate: real JPEG

integer DQTs → MozJPEG/libjpeg → file bytes

common decoder → MS-SSIM / PSNR

bpp = 8 × complete file bytes ÷ source pixels